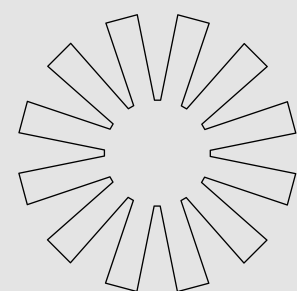
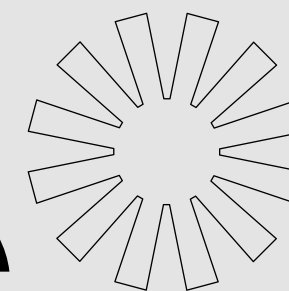




WS PROJECT 1



MUSIC TRENDS



ÂNGELA RIBEIRO (109061) | ANTÓNIO ALBERTO (114622) | HUGO CASTRO (113889)

14 APRIL 2026

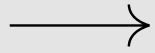


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02



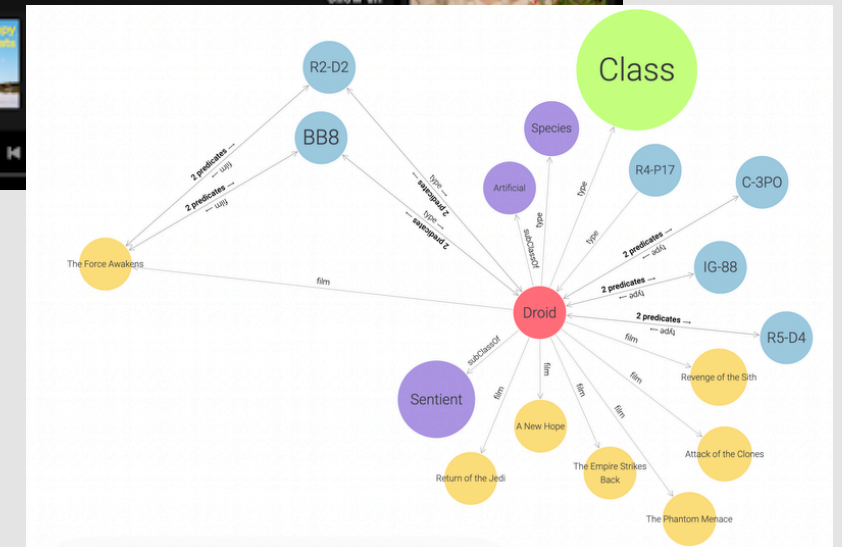
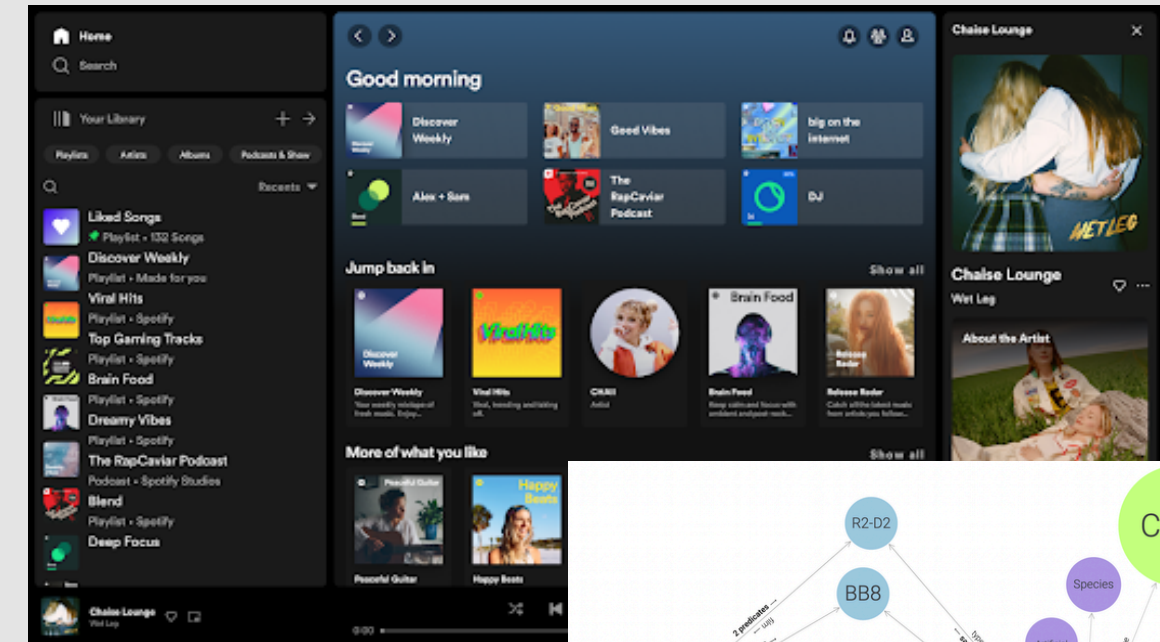
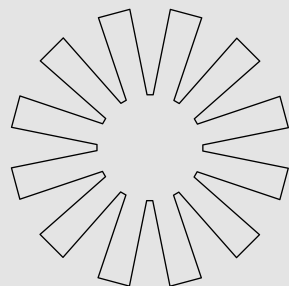
MOTIVATION AND GOALS

Why this topic?

- Music data contains rich relationships between songs, artists, genres, collaborations, and charts
- These relationships are well suited to graph-based representation

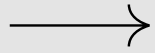
Project objectives

- Build an RDF dataset from music-related sources
- Store and query the data using GraphDB and SPARQL
- Develop a Django web application for exploration and management





03



MUSIC TRENDS

DATA SOURCES

BILLBOARD HOT 100

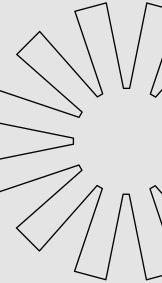
- Weekly chart records since 2004
- Fields: song, artist, rank, weeks on chart, date

<https://www.kaggle.com/datasets/ludmin/billboard>

SPOTIFY TRACKS DETAILED DATASET

- Audio features and semantic enrichment
- Fields: energy, danceability, valence, album, genre, tempo, etc.

<https://www.kaggle.com/datasets/saichaitanyareddyai/spotify-tracks-dataset-audio-features>



Billboard Hot 100 & more

A complete record of the Billboard Hot 100 from its inception to the present day



[Data Card](#) [Code \(1\)](#) [Discussion \(0\)](#) [Suggestions \(0\)](#)

About Dataset

 **Billboard Historical Charts (Auto-Updated Weekly)**

A curated, long-horizon collection of major **Billboard music charts** — from their beginnings (as far back as publicly accessible) **up to the present**. Data is refreshed **every Wednesday at 02:00 (server time)** so you always have up-to-date rankings for analysis, trending, and music data science projects.

 **Included Chart Files**

Each file is a time series of weekly chart positions:

File	Chart	Type
billboard200.csv	Billboard 200	Albums

Usability ⓘ
10.00

License
[CC BY-NC-SA 4.0](#)

Expected update frequency
Weekly

Tags

[Arts and Entertainment](#)

[Music](#) [Tabular](#)

[Categorical](#) [Global](#)

Spotify Tracks Dataset | Audio Features

114K Spotify tracks with audio features for EDA & Machine Learning



[Data Card](#) [Code \(2\)](#) [Discussion \(0\)](#) [Suggestions \(0\)](#)

About Dataset

 **Spotify Tracks Dataset — 114K Songs Across 114 Genres**

A comprehensive collection of **Spotify tracks** enriched with audio features, ideal for music analysis, genre classification, and building recommendation systems.

 **About This Dataset**

This dataset contains **114,000 Spotify tracks** spanning **114 unique music genres**, collected via the **Spotify Web API**. Each track comes with a rich set of audio features computed by Spotify's internal audio analysis engine — making it perfect for anyone interested in the science of music.

Whether you're a beginner practicing EDA or an advanced practitioner building ML models, this dataset offers meaningful signals and real-world complexity.

Usability ⓘ
7.06

License
[CC0: Public Domain](#)

Expected update frequency
Not specified

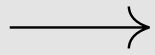
Tags

[Music](#) [Data Analytics](#)

[Exploratory Data Analysis](#)

[Feature Engineering](#)

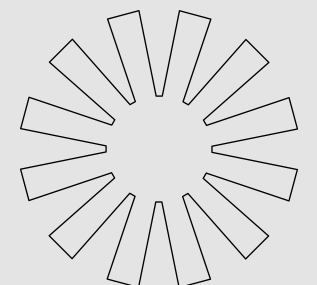
[Audio Classification](#)



MAIN DIFFICULTIES

- Combine the datasets
- Normalize the artists

Date	Song	Artist
2004-10-20	Goodies	Ciara Featuring Petey Pablo
2004-10-20	Drop It Like It's Hot	Snoop Dogg Feat. Pharrell
2004-10-20	Lean Back	Terror Squad
2004-10-20	Welcome To My Life	Simple Plan
2004-10-20	American Idiot	Green Day
2004-10-20	Tilt Ya Head Back	Nelly & Christina Aguilera





05

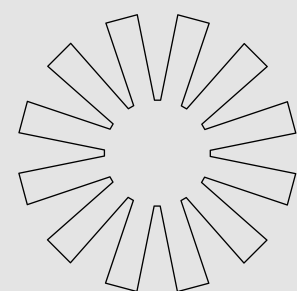


MUSIC TRENDS

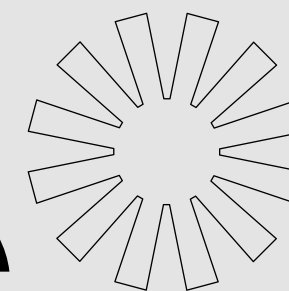
LIVE
DEMO



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